

Coding & Solution

Date	07 July 2026
Team ID	SWTID-2026-4350
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

- Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- The solution should address the core problem statement and implement the key functional requirements.
- Include all source files, configuration files, and setup instructions needed to run the application.
- Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/SaiRajesh01/Credit-Card-Approval-Prediction_smart_bridge
Programming Language(s)	Python, HTML5, CSS3, JavaScript, SQL
Framework(s) Used	Flask, Flask-SQLAlchemy, Waitress, XGBoost, Scikit-learn
Key Features Implemented	User Management (Admin & Analyst) • Applicant Information Management • Credit History Storage • Loan Approval Prediction using Machine Learning
Pending / Incomplete Features	• User Authentication (Login/Logout) • Email Notifications • File Upload for Bulk Predictions • Model Retraining Interface
Setup / Run Instructions	Install Python 3.10 or above. 2. Install required libraries using <code>pip install -r requirements.txt</code> . 3. Configure the database in <code>config.py</code> . 4. Place the trained ML model in the appropriate directory. 5. Run <code>python app.py</code> . 6. Open <code>http://localhost:5000</code> in a web browser. 7. Enter applicant information and submit the prediction form. 8. View prediction results and dashboard statistics.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	YES
2	Meaningful variable and function names are used	YES
3	Code includes comments / documentation where necessary	YES
4	Error handling is implemented for critical operations	YES
5	The application runs without critical errors	YES
6	Code is committed to a version control repository	YES

Additional Notes / Comments:

- ☐ The project follows a modular architecture using the Flask framework.
- ☐ SQLAlchemy ORM is used for efficient database management.
- ☐ Machine learning predictions are integrated through a dedicated prediction module (CreditPredictor).
- ☐ The system stores applicant details, prediction history, and model metadata in the database.